Complex Eigenvalues

We've looked at distinct real Eigenvalues and equal real Eigenvalues of multiplicity two and multiplicity three. Now we want to look at **complex Eigenvalues**, which are Eigenvalues that are complex numbers.

We'll find complex Eigenvalues $\lambda_1 = \alpha + \beta i$ and $\lambda_2 = \alpha - \beta i$ using the quadratic formula to solve the characteristic equation, and then we'll find the associated Eigenvectors in the same way we did before, by solving the system of equations we get from $|A - \lambda I| = \overrightarrow{O}$.

The Eigenvectors $\overrightarrow{k_i}$ that we find still get plugged into our solutions

$$\overrightarrow{x_i} = \overrightarrow{k_i} e^{\lambda_i t}$$

And then the general solution will be given in the same form we're used to seeing,

$$\overrightarrow{x} = c_1\overrightarrow{x_1} + c_2\overrightarrow{x_2} + \ldots + c_n\overrightarrow{x_n}$$

The Eigenvalues $\lambda_1 = \alpha + \beta i$ and $\lambda_2 = \alpha - \beta i$ are conjugates, and the Eigenvectors $\overrightarrow{k_1}$ and $\overrightarrow{k_2}$ are conjugates.

Let's do an example where we use Eigenvalue/Eigenvector pairs for a set of complex conjugate Eigenvalues in order to solve a system of differential equations.

Example

Solve the system of differential equations.

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$$x_1' = 4x_1 + 5x_2$$

$$x_2' = -2x_1 + 6x_2$$

The coefficient matrix is

$$A = \begin{bmatrix} 4 & 5 \\ -2 & 6 \end{bmatrix}$$

and the matrix $A - \lambda I$ is

$$A - \lambda I = \begin{bmatrix} 4 & 5 \\ -2 & 6 \end{bmatrix} - \lambda \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

$$A - \lambda I = \begin{bmatrix} 4 & 5 \\ -2 & 6 \end{bmatrix} - \begin{bmatrix} \lambda & 0 \\ 0 & \lambda \end{bmatrix}$$

$$A - \lambda I = \begin{bmatrix} 4 - \lambda & 5 \\ -2 & 6 - \lambda \end{bmatrix}$$

Its determinant is

$$\begin{vmatrix} 4 - \lambda & 5 \\ -2 & 6 - \lambda \end{vmatrix} = (4 - \lambda)(6 - \lambda) - (5)(-2)$$

$$\begin{vmatrix} 4 - \lambda & 5 \\ -2 & 6 - \lambda \end{vmatrix} = 24 - 4\lambda - 6\lambda + \lambda^2 + 10$$

$$\begin{vmatrix} 4 - \lambda & 5 \\ -2 & 6 - \lambda \end{vmatrix} = \lambda^2 - 10\lambda + 34$$

So the characteristic equation is

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$$\lambda^2 - 10\lambda + 34 = 0$$

Using the quadratic formula, the roots are

$$\lambda = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$\lambda = \frac{-(-10) \pm \sqrt{(-10)^2 - 4(1)(34)}}{2(1)}$$

$$\lambda = \frac{10 \pm \sqrt{-36}}{2}$$

$$\lambda = \frac{10 \pm 6i}{2}$$

$$\lambda = 5 \pm 3i$$

Then for these complex conjugate Eigenvalues, $\lambda_1 = 5 + 3i$ and $\lambda_2 = 5 - 3i$, we find

$$A - (5 + 3i)I = \begin{bmatrix} 4 - (5 + 3i) & 5 \\ -2 & 6 - (5 + 3i) \end{bmatrix}$$

$$A - (5 + 3i)I = \begin{bmatrix} -1 - 3i & 5 \\ -2 & 1 - 3i \end{bmatrix}$$

and

$$A - (5 - 3i)I = \begin{bmatrix} 4 - (5 - 3i) & 5 \\ -2 & 6 - (5 - 3i) \end{bmatrix}$$

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$$A-(5-3i)I=\begin{bmatrix}-1+3i & 5\\ -2 & 1+3i\end{bmatrix}$$

If we turn matrices back into systems of equations, we get

$$(-1-3i)k_1+(5)k_2=0$$

$$(-2)k_1+(1-3i)k_2=0$$

and

$$(-1+3i)k_1+(5)k_2=0$$

$$(-2)k_1+(1+3i)k_2=0$$

From the first equation in the first system for $\lambda_1=5+3i$, we find

$$k_2=\left(\frac{1}{5}+\frac{3}{5}i\right)k_1$$

Choosing $k_1=5$ gives $k_2=1+3i$, which leads us to the Eigenvector

$$\overrightarrow{k_1}=\begin{bmatrix}5\\ 1+3i\end{bmatrix}$$

and the corresponding solution vector

$$\overrightarrow{x_1}=\overrightarrow{k_1}e^{\lambda_1 t}$$

$$\overrightarrow{x_1}=\begin{bmatrix}5\\ 1+3i\end{bmatrix}e^{(5+3i)t}$$

In the same way, from the first equation in the second system for $\lambda_2=5-3i$, we find

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$$k_2 = \left(\frac{1}{5} - \frac{3}{5}i\right) k_1$$

Choosing $k_1 = 5$ gives $k_2 = 1 - 3i$, which leads us to the Eigenvector

$$\overrightarrow{k_2} = \begin{bmatrix} 5 \\ 1 - 3i \end{bmatrix}$$

and the corresponding solution vector

$$\overrightarrow{x_2} = \overrightarrow{k_2} e^{\lambda_2 t}$$

$$\overrightarrow{x_2} = \begin{bmatrix} 5 \\ 1 - 3i \end{bmatrix} e^{(5-3i)t}$$

Therefore, the general solution will be

$$\overrightarrow{x} = c_1\overrightarrow{x_1} + c_2\overrightarrow{x_2}$$

$$\overrightarrow{x} = c_1 \begin{bmatrix} 5 \\ 1 + 3i \end{bmatrix} e^{(5+3i)t} + c_2 \begin{bmatrix} 5 \\ 1 - 3i \end{bmatrix} e^{(5-3i)t}$$

The solution in real terms

We can convert the general solution from complex numbers into real numbers. When we have two complex Eigenvalues, $\lambda_1 = \alpha + \beta i$ and $\lambda_2 = \alpha - \beta i$, we only need to consider one Eigenvector. For $\lambda_1 = \alpha + \beta i$, the real vector solutions will be

$$\overrightarrow{x_1} = [\overrightarrow{b_1}\cos(\beta t) - \overrightarrow{b_2}\sin(\beta t)]e^{\alpha t}$$

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$$\overrightarrow{x_2} = [\overrightarrow{b_2}\cos(\beta t) + \overrightarrow{b_1}\sin(\beta t)]e^{\alpha t}$$

where $\overrightarrow{b_1}$ is the real part of the Eigenvector $\overrightarrow{k_1}$, $\overrightarrow{b_1} = \text{Re}(\overrightarrow{k_1})$, and where $\overrightarrow{b_2}$ is the imaginary part of the Eigenvector $\overrightarrow{k_1}$, $\overrightarrow{b_2} = \text{Im}(\overrightarrow{k_1})$. The complex conjugate Eigenvalue $\lambda_2 = \alpha - \beta i$ gives the same values for α and β, and therefore the same two solutions $\overrightarrow{x_1}$ and $\overrightarrow{x_2}$.

Let's extend the last example, to change the solution we found into real terms instead of complex terms.

Example (cont'd)

Convert the general solution into real terms.

$$\overrightarrow{x} = c_1\begin{bmatrix} 5 \\ 1+3i \end{bmatrix}e^{(5+3i)t} + c_2\begin{bmatrix} 5 \\ 1-3i \end{bmatrix}e^{(5-3i)t}$$

The Eigenvector $\overrightarrow{k_1}$ can be rewritten as separate real and imaginary parts.

$$\overrightarrow{k_1} = \begin{bmatrix} 5 \\ 1+3i \end{bmatrix}$$

$$\overrightarrow{k_1} = \begin{bmatrix} 5 \\ 1 \end{bmatrix} + i\begin{bmatrix} 0 \\ 3 \end{bmatrix}$$

Therefore, the vectors $\overrightarrow{b_1} = \text{Re}(\overrightarrow{k_1})$ and $\overrightarrow{b_2} = \text{Im}(\overrightarrow{k_1})$ are

$$\overrightarrow{b_1} = \begin{bmatrix} 5 \\ 1 \end{bmatrix}$$

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$$\vec{b_2} = \begin{bmatrix} 0 \\ 3 \end{bmatrix}$$

So the first solution will be

$$\vec{x_1} = [\vec{b_1}\cos(\beta t) - \vec{b_2}\sin(\beta t)]e^{\alpha t}$$

$$\vec{x_1} = \left(\begin{bmatrix} 5 \\ 1 \end{bmatrix} \cos(3t) - \begin{bmatrix} 0 \\ 3 \end{bmatrix} \sin(3t) \right) e^{5t}$$

$$\vec{x_1} = \left(\begin{bmatrix} 5\cos(3t) \\ \cos(3t) \end{bmatrix} - \begin{bmatrix} 0 \\ 3\sin(3t) \end{bmatrix} \right) e^{5t}$$

$$\vec{x_1} = \begin{bmatrix} 5\cos(3t) \\ \cos(3t) - 3\sin(3t) \end{bmatrix} e^{5t}$$

The second solution will be

$$\vec{x_2} = [\vec{b_2}\cos(\beta t) + \vec{b_1}\sin(\beta t)]e^{\alpha t}$$

$$\vec{x_2} = \left(\begin{bmatrix} 0 \\ 3 \end{bmatrix} \cos(3t) + \begin{bmatrix} 5 \\ 1 \end{bmatrix} \sin(3t) \right) e^{5t}$$

$$\vec{x_2} = \left(\begin{bmatrix} 0 \\ 3\cos(3t) \end{bmatrix} + \begin{bmatrix} 5\sin(3t) \\ \sin(3t) \end{bmatrix} \right) e^{5t}$$

$$\vec{x_2} = \begin{bmatrix} 5\sin(3t) \\ 3\cos(3t) + \sin(3t) \end{bmatrix} e^{5t}$$

Therefore, the general solution in real terms is given by

$$\vec{x} = c_1 \begin{bmatrix} 5\cos(3t) \\ \cos(3t) - 3\sin(3t) \end{bmatrix} e^{5t} + c_2 \begin{bmatrix} 5\sin(3t) \\ 3\cos(3t) + \sin(3t) \end{bmatrix} e^{5t}$$

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